

**SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ**

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0510

Slot A

COURSE OUTCOME COVERED

CO3: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)

Assessment Tool Weightage: 10%

QUESTIONS:50

Total Marks 50

Duration :60 Minutes

Annexure A MCQ

Code of Conduct:

I P. Akhila (192425225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Akhila

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect	
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers	
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts	
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time	
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions	

1. Preprocessing – Normalization & Scaling

A 12-bit image has pixel value $r = 3000$.

The image is normalized to $[0,1]$ and then scaled to 8-bit $[0,255]$.

What is the final pixel value?

- A) 150
- B) 187
- C) 200
- D) 225

Ans: B) 187

2. Image Storage Calculation

An RGB image of size 2048×1024 uses 8 bits per channel.

What is the storage in MB?

- A) 6 MB
- B) 8 MB
- C) 12 MB
- D) 24 MB

Ans: A) 6 MB

3. Sobel Edge Magnitude

At a pixel: ($G_x = 10$, $G_y = 24$)

Compute the gradient magnitude (approx).

- A) 20
- B) 25
- C) 26
- D) 30

Ans: C) 26

4. Sobel Edge Direction

For the same ($G_x = 10$, $G_y = 24$), compute edge direction in degrees.

- A) 22°
- B) 67°
- C) 78°
- D) 90°

Ans: B) 67°

5. Harris Corner Response $\det(M)$

$= 500$, $\text{trace}(M) = 40$, $k = 0.04$

Compute Harris response R .

- A) 440
- B) 460
- C) 500
- D) 520

Ans: A) 440

6. Harris Corner vs Edge Classification

Harris matrix eigenvalues: $\lambda_1 = 100$, $\lambda_2 = 5$

The pixel corresponds to:

- A) Flat region
- B) Edge
- C) Corner
- D) Noise

Ans: B) Edge

7. Hough Transform – Line

Parameter A line in Hough space: $\rho = 50$, $\theta = 90^\circ$

Which line is represented?

- A) Horizontal line $y = 50$
- B) Vertical line $x = 50$
- C) Diagonal
- D) Random line

Ans: B) Vertical line $x = 50$

8. SIFT Scale Adjustment

Original keypoint $\sigma = 1.2$, image scaled by factor 2.

New keypoint scale = ?

- A) 1.2
- B) 2.4
- C) 0.6
- D) 3.2

Ans: B) 2.4

9. PCA Variance Calculation

Eigenvalues of a dataset: 50, 25, 15, 10

How many principal components retain $\geq 90\%$ variance?

- A) 2
- B) 3
- C) 4
- D) 5

Ans: B) 3

10. PCA Dimensionality Reduction Impact

Original feature vector: 200 dimensions

Reduced via PCA: 40 dimensions

Percentage reduction = ?

A) 70%

B) 75%

C) 80%

Ans: C) 80%

11. Preprocessing – Normalization Scenario

A drone captures an 12-bit image with pixel value $r = 2500$.

Before feature extraction, it is normalized to $[0,1]$.

What is the normalized value?

A) 0.55

B) 0.61

C) 0.65

D) 0.72

Ans: B) 0.61

12. Image Storage Scenario

A grayscale image: 2048×1024 , 16-bit per pixel. You convert it to 8-bit for faster processing.

What is the percentage reduction in memory usage?

A) 25%

B) 50%

C) 60%

D) 75%

Ans: B) 50%

13. Edge Detection Scenario

A Sobel operator detects gradients at a pixel: ($G_x = 12$, $G_y = 16$).

Compute the gradient magnitude.

A) 18

B) 20

C) 24

D) 28

Ans: B) 20

14. Edge Direction Scenario

For the same pixel above, compute gradient direction θ .

A) 36.9°

B) 45°

C) 53.1°

D) 60°

Ans: A) 36.9°

15. Harris Corner Detection Scenario

Region M parameters: $\det(M) = 600$, $\text{trace}(M) = 30$, $k = 0.05$

Compute Harris response R.

A) 540

B) 555

C) 570

D) 600

Ans: B) 555

16. Harris Corner Classification Scenario

Eigenvalues at a pixel: $\lambda_1 = 100$, $\lambda_2 = 90$

The pixel is most likely:

- A) Flat region
- B) Edge
- C) Corner
- D) Noise

Ans: C) Corner

17. Hough Transform Scenario

Two points in an image produce intersecting curves in Hough space at ($\rho = 80$, $\theta = 60^\circ$).

What does this indicate?

- A) Detected corner
- B) Line in image space
- C) Circular feature
- D) Noise

Ans: B) Line in image space

18. SIFT Scale Scenario

A SIFT keypoint appears at scale $\sigma = 2.0$.

The image is downsampled by a factor of 0.5.

What is the new keypoint scale?

- A) 0.5
- B) 1.0
- C) 2.0

D) 4.0

Ans: B) 1.0

19. PCA Variance Scenario

A dataset has eigenvalues: 70, 15, 10, 5

How many principal components retain $\geq 90\%$ variance?

A) 2

B) 3

C) 4

Ans: B) 3

20. PCA Dimensionality Reduction Scenario

Original feature vector: 150 dimensions

Reduced via PCA: 25 dimensions

What is the percentage reduction?

A) 70%

B) 75%

C) 83%

D) 85%

Ans: C) 83%

21. Preprocessing – Normalization & Scaling

A 12-bit satellite image has a pixel intensity $r = 3500$.

It is normalized to $[0,1]$ and then scaled to 8-bit $[0,255]$ for processing.

Question: What is the final 8-bit pixel value?

A) 190

- B) 210
- C) 218
- D) 225

Ans: C) 218

22. Image Storage Analysis

A 1024×1024 RGB image with 8-bit per channel is converted to grayscale 8-bit.

Question: What is the percentage reduction in storage?

- A) 33%
- B) 50%
- C) 66%
- D) 75%

Ans: C) 66%

23. Edge Detection – Gradient Magnitude A pixel has Sobel gradients: ($G_x = 14$, $G_y = 48$)

Question: Compute the gradient magnitude.

- A) 46
- B) 50
- C) 50.5
- D) 50.99

Ans: B) 50

24. Edge Direction

For the same pixel: ($G_x = 14$, $G_y = 48$)

Question: Compute edge orientation in degrees.

- A) 70°
- B) 73°
- C) 75°
- D) 77°

Ans: C) 75°

25. Harris Corner Response

Harris parameters: $\det(M) = 400$, $\text{trace}(M) = 35$, $k = 0.05$

Question: Compute Harris response R.

- A) 320
- B) 350
- C) 378
- D) 400

Ans: No exact option

26. Harris Corner Classification

Eigenvalues of M: $\lambda_1 = 130$, $\lambda_2 = 120$

Question: Pixel type?

- A) Flat
- B) Edge
- C) Corner
- D) Noise

Ans: C) Corner

27. Hough Transform – Line Detection

Two points in image space generate intersecting curves in Hough space at $(\rho = 100, \theta = 45^\circ)$.

Question: What does this indicate in the image?

- A) A line passes through both points
- B) A corner formed by lines
- C) Circular feature
- D) Random noise

Ans: A) A line passes through both points

28. SIFT Scale Scenario

A SIFT keypoint at scale $\sigma = 1.5$ is observed. The image is scaled by $0.25\times$.

Question: New keypoint scale?

- A) 0.375
- B) 0.75
- C) 1.5
- D) 6

Ans: A) 0.375

29. PCA Variance Retention

Eigenvalues: 60, 20, 10, 5, 5

Question: Minimum principal components to retain $\geq 90\%$ variance?

- A) 2
- B) 3
- C) 4
- D) 5

Ans: B) 3

30. PCA Dimensionality Reduction

Original vector dimension = 180

After PCA = 30 components

Question: Percentage reduction?

- A) 70%
- B) 75%
- C) 83%
- D) 85%

Ans: C) 83%

31. Preprocessing Method Selection

A remote sensing image contains high sensor noise and uneven illumination. You need to enhance edges for feature extraction.

Which preprocessing combination is most appropriate?

- A) Histogram equalization + median filtering
- B) Gaussian smoothing + normalization
- C) Median filtering + adaptive histogram equalization
- D) Low-pass filtering + quantization

Ans: C) Median filtering + adaptive histogram equalization

32. Image Representation Trade-Off

A 3D medical scan (1024×1024) can be stored as 16-bit grayscale or 8-bit with histogram equalization.

Which representation maximizes detail and reduces storage without losing features?

- A) 16-bit grayscale
- B) 8-bit grayscale with HE
- C) 8-bit grayscale without HE
- D) RGB 24-bit

Ans: B) 8-bit grayscale with HE

33. Edge Detection Algorithm Choice

You are analyzing a satellite image with low contrast edges and some noise.

Which edge detector is most suitable?

- A) Sobel
- B) Prewitt
- C) Canny
- D) Roberts

Ans: C) Canny

34. Harris vs FAST Corner Selection

You need to detect corners for real-time object tracking.

Which method is preferred for speed without losing accuracy?

- A) Harris
- B) FAST
- C) SIFT
- D) Hough Transform

Ans: B) FAST

35. Hough Transform Line Detection Evaluation

Two line detection methods are applied on a road image:

- Method 1: ρ -resolution = 1 pixel, θ -resolution = 1°
- Method 2: ρ -resolution = 5 pixels, θ -resolution = 5°

Which method balances accuracy and computational cost best?

- A) Method 1
- B) Method 2
- C) Both equal

D) Use Canny edges only

Ans: B) Method 2

36.. SIFT vs SURF Evaluation

You are matching keypoints between two images with scale and rotation differences.

Which method ensures robust matching with lower computation?

A) SIFT

B) SURF

C) Harris + SSD

D) Hough Transform

Ans: B) SURF

37. PCA Component Selection

A dataset has eigenvalues: 70, 20, 10, 5, 5

To retain $\geq 95\%$ variance for classification:

A) 2 components

B) 3 components

C) 4 components

D) 5 components

Ans: C) 4 components

38. Feature Extraction Evaluation

You have a noisy industrial inspection image. You want to detect straight edges of objects.

Which combination is optimal?

A) Sobel + Hough Transform

B) Canny + Harris Corner

C) Median Filter + SIFT

D) Gaussian Blur + PCA

Ans: A) Sobel + Hough Transform

39. Preprocessing for SIFT Matching

Two aerial images have different brightness and contrast.

What preprocessing ensures scale-invariant feature matching works best?

A) Histogram equalization + normalization

B) Median filtering + low-pass filter

C) Gaussian smoothing only

D) No preprocessing

Ans: A) Histogram equalization + normalization

40. PCA Dimensionality Evaluation

A feature vector of 200 dimensions is reduced using PCA to 30 components, retaining $\approx 90\%$ variance.

You need to classify images quickly.

Which evaluation is correct?

A) Reduction sufficient for speed and accuracy

B) Reduce to 10 components for maximum speed

C) Keep 200 components for highest accuracy

D) Use SIFT only

Ans: A) Reduction sufficient for speed and accuracy

41. Preprocessing – Normalization & Decision

A 12-bit satellite image has a pixel $r = 3200$.

You normalize it to $[0,1]$ and scale to 8-bit $[0,255]$.

Question: What is the final pixel value?

- A) 198
- B) 199
- C) 200

Ans: B) 199

42. Image Storage Trade-Off

A 1024×1024 RGB image (8-bit per channel) is converted to grayscale 8-bit to save storage.

Question: What is the percentage reduction?

- A) 33%
- B) 50%
- C) 66.7%
- D) 75%

Ans: C) 66.7%

43. Sobel Edge Magnitude Evaluation

A pixel has gradients ($G_x = 12$, $G_y = 16$).

Question: Compute gradient magnitude and evaluate which pixel is stronger than another pixel with ($G_x = 10$, $G_y = 18$).

- A) 20; weaker
- B) 20; stronger
- C) 21; weaker
- D) 21; stronger

Ans: A) 20; weaker

44. Harris Corner Response Analysis

$\det(M)=400$, $\text{trace}(M)=30$, $k=0.04$

Question: Compute R and evaluate whether it's more likely a corner than a pixel with

$\det(M)=350$, $\text{trace}(M)=25$, $k=0.04$.

- A) $R=364$; less likely
- B) $R=364$; more likely
- C) $R=362$; less likely
- D) $R=362$; more likely

Ans: B) $R=364$; more likely

45. Harris Eigenvalue Evaluation

Eigenvalues of M at a pixel: $\lambda_1=120$, $\lambda_2=80$

Question: Classify pixel as flat, edge, or corner numerically.

- A) Flat
- B) Edge
- C) Corner
- D) Noise

Ans: C) Corner

46. Hough Transform Voting Numerical

Two points in image space: $(100,50)$ and $(200,100)$

Each generates sinusoidal curves in Hough space. They intersect at $(\rho=70, \theta=45^\circ)$.

Question: What does this indicate?

- A) Line passes through both points
- B) Corner
- C) Noise
- D) Circle

Ans: A) Line passes through both points

47. SIFT Keypoint Scale Numerical Evaluation

Original keypoint $\sigma=1.6$; image scaled $0.5\times$

Question: New σ value?

- A) 0.5
- B) 0.8
- C) 1.6
- D) 3.2

Ans: B) 0.8

48. PCA Variance Contribution Evaluation

Eigenvalues: 60, 20, 10, 5, 5

Question: Minimum principal components to retain $\geq 90\%$ variance?

- A) 2
- B) 3
- C) 4
- D) 5

Ans: B) 3

49. PCA Dimensionality Reduction Numerical Evaluation

Original feature dimension = 200

After PCA = 30 components

Question: Evaluate percentage reduction.

- A) 70%
- B) 75%
- C) 80%
- D) 85%

Ans: D) 85%

50. Preprocessing Impact Evaluation

Two 8-bit images:

- Image A has uniform illumination
- Image B has low contrast and brightness differences

You apply histogram equalization to both.

Question: Evaluate which image benefits more numerically in contrast enhancement.

- A) Image A
- B) Image B
- C) Both equal
- D) Neither

Ans: B) Image B